

Clinical Study Summary Report (CSSR)

Document Status: Draft | Version: 1.0 | Date: 02-Apr-2026

1. Administrative & Study Identification

Full Study Title: Non-interventional study on Guideline Directed Medical Therapy for patients with heart failure (HF) in Germany - PHRASE-HF **Short Title/Acronym:** PHRASE-HF **Protocol Number:** D1699R00045 **NCT Number:** NCT06675552 **EudraCT Identifier:** N/A **CTIS Identifier:** N/A **Sponsor Name:** AstraZeneca **Verification:** Verified 01-Feb-2026 by AstraZeneca **Investigational Product:** N/A (Observational study evaluating Guideline Directed Medical Therapy: ARNI/ACE-I/ARB, BB, MRA, and SGLT2i; Study drug: -) **Phase of Development:** N/A (Observational / Non-interventional) **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the use of Guideline Directed Medical Therapy (GDMT) at hospital discharge, the translation of in-hospital implementation, and the possible maximization of phenotype-based GDMT into real-world routine outpatient care. **Secondary Objectives:** To evaluate heart failure symptoms, patient-reported outcomes (PROs), clinically relevant outcomes (e.g., rehospitalization, mortality), and the use of diuretics and concomitant drug classes.

2.2 Background & Rationale To address the clinical and economic burden of Heart Disease, specifically Heart Failure with reduced Ejection Fraction (HFrEF), which represents approximately 50% of the HF patient population. Despite current ESC guideline recommendations that combine four drug classes with proven prognostic benefit, hospitalization and mortality rates remain especially high. This study generates real-world evidence to observe how in-hospital GDMT initiation translates into long-term outpatient adherence.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational **Allocation:** N/A **Control Type:** Single-arm (Non-interventional) **Intervention Model:** - **Blinding (Masking):** - (Open-label) **Primary Purpose:** - **Endpoint Classification:** - **Randomization:** N/A

3.2 Planned Enrollment & Duration Estimated Enrollment (Target \$N\$): 438 patients **Number of Sites:** Participating sites trained within the Helios-GDMT-program in Germany **Study Start Date:** 15-Nov-2024 **Estimated Primary Completion Date:** 30-Jun-2027 **Estimated Study Completion Date:** 30-Jun-2027

4. Subject Selection (Eligibility Criteria)

4.1 General Demographics Age: 18 Years - 130 Years Sex: All
Healthy Volunteers: No

4.2 Inclusion Criteria 1. Age 18 Years - 130 Years at the time of signing the informed consent. 2. Hospitalized in a participating site and receiving full inpatient treatment (at least 24h hospital stay). 3. Diagnosis of HFrEF according to the current guidelines of the European Society of Cardiology (ESC) with a left ventricular EF of $\leq 40\%$. 4. Treated with a maximum of 2 of the indicated drug classes (ACE-I/ARNI/ARB, BB, MRA, SGLT2i) according to guideline recommendation at admission.

4.3 Exclusion Criteria 1. Lack of signed and dated written informed consent prior to enrolment. 2. Concurrent participation in an interventional clinical trial that dictates specific HF therapies. 3. Patients not meeting the strict HFrEF diagnostic criteria.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Routine clinical care based on standard GDMT (ARNI/ACE-I/ARB, BB, MRA, and SGLT2i) titrated at the physician's discretion. **Route of Administration:** Oral (Standard disease management) **Duration of Treatment:** Real-world follow-up from hospital discharge until study completion in June 2027.

5.2 Concomitant & Prohibited Therapy Permitted: Use of diuretics and any other concomitant drug classes required for co-morbidities as per routine care. **Prohibited:** N/A (Observational study).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Informed Consent, Medical History, Confirmation of LVEF $\leq 40\%$
Baseline	Day 0	Index Hospitalization, Assessment of GDMT status at admission vs. discharge
Interim	Post-discharge	Routine outpatient care: PRO collection, HF symptom assessment, GDMT optimization tracking
End of Study	30-Jun-2027	Final assessment of clinically relevant outcomes (rehospitalization, mortality)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: The evaluation of the use of GDMT at hospital discharge and its translation and maximization into routine outpatient care. **Measurement Method:** Retrospective and prospective review of electronic health records (EHR) and prescription databases.

7.2 Secondary & Exploratory Endpoints Patient-reported outcomes (PROs). Clinically relevant outcomes, specifically rehospitalization rates and mortality. Changes in HF symptoms and the use of diuretics/concomitant drug classes.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Extracted via standard real-world data collection and patient medical records. **Serious Adverse Events (SAEs):** Major clinical events (e.g., rehospitalization, mortality) captured primarily as clinical outcome endpoints. **Data Monitoring Committee (DMC):** Supervised by AstraZeneca's observational research governance team.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Total study population (All enrolled and eligible patients); subgroups of interest defined by exploratory objectives. **Sample Size Justification:** $N = 438$ provides sufficient real-world volume to detect trends in outpatient GDMT maximization. **Handling of Missing Data:** Standard handling for observational registries (e.g., Multiple Imputation or complete case analysis depending on the outcome variable).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Pharmacoepidemiology Practices (GPP) and the Declaration of Helsinki. **Monitoring Plan:** Risk-based monitoring focusing on data completeness at Helios-GDMT-program trained sites. **Audit Trail:** Secure electronic data capture (EDC) with tracking of clinical pathway adherence and automated data clarification forms.

11. Study Identifiers & Governance

Primary ID: D1699R00045 **Registry Number:** NCT06675552 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** PHRASE-HF **Official Regulatory Title:** Non-interventional study on Guideline Directed Medical Therapy for patients with heart failure (HF) in Germany - PHRASE-HF

12. Clinical Framework (Disease Specific)

Primary Condition: Heart Failure with reduced Ejection Fraction (HFrEF) **Medical Condition:** Heart Disease **Diagnosis Threshold:** LVEF $\leq 40\%$ **Medical Methodology:** Guideline Directed Medical Therapy (GDMT) **Optimization Target:** 4-pillar GDMT implementation (ARNI/ACE-I/ARB, BB, MRA, SGLT2i)

13. Study Procedures & Methodology

Management Pathway: Inpatient to outpatient transition pathway for HFrEF **Education Protocol (HCP):** Helios-GDMT-program site training **Education Protocol (Patient):** Real-world outpatient disease counseling **Quality Audit Mechanism:** Tracking of in-hospital prescription translations to real-world routine outpatient care

14. Observation & Follow-Up Schedule

Total Duration: Follow-up through June 2027 **Onsite Visit Cadence:** Index hospitalization discharge and routine clinical visits **Remote Monitoring Frequency:** Periodic Electronic Health Record (EHR) review **Checklist Review Interval:** According to routine standard of care

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				